Pair of Linear Equations in Two Variables (Set B) — MCQ Worksheet

Mathematics • Chapter 3 • *25 Questions, 1 mark each — Total: 25 marks*

Instructions: *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

Q1. The pair x + y = 6, x – y = 4 has solution:

(A) (5, 1) (B) (1, 5)
(C) (4, 2) (D) (2, 4)

Q2. For the lines 2x + 3y = 5 and 4x + 6y = 10, the system has:

(A) Unique solution (B) No solution
(C) Infinitely many solutions (D) Two solutions

Q3. For 3x + 2y = 5 and 6x + 4y = 8, the system has:

(A) Unique solution (B) No solution
(C) Infinitely many solutions (D) Two solutions

Q4. Lines are intersecting (have unique solution) when a_1/a_2:

(A) $= b_1/b_2$ (B) $\neq b_1/b_2$
(C) $= b_1/b_2 = c_1/c_2$ (D) $= b_1/b_2 \neq c_1/c_2$

Q5. Solution of x + 3y = 11, x – 3y = –1 is:

(A) (5, 2) (B) (2, 5)
(C) (5, 2/3) (D) (2, 5/3)

Q6. If 2x + 3y = 8 and 4x + 6y = 16, the pair is:

(A) Consistent with unique solution (B) Inconsistent
(C) Consistent with infinitely many solutions (D) Has no real solution

Q7. A two-digit number, when the digits are reversed, becomes a number greater than the original by 27. If sum of digits is 9, the number is:

(A) 36 (B) 63
(C) 45 (D) 54

Q8. Sum of two numbers is 50 and their difference is 10. The numbers are:

(A) 20, 30 (B) 25, 25
(C) 15, 35 (D) 30, 20

Q9. Cost of 2 books and 3 pens is Rs 80 and 3 books and 2 pens is Rs 70. Cost of 1 book is:

(A) Rs 10 (B) Rs 15
(C) Rs 20 (D) Rs 25

Q10. If we add 1 to numerator and subtract 1 from denominator, fraction becomes 1. If we add 1 only to denominator, it becomes 1/2. The fraction is:

(A) 3/5 (B) 4/7
(C) 5/7 (D) 3/7

Q11. Equation y = mx + c represents:

(A) A circle (B) A parabola
(C) A straight line (D) A hyperbola

Q12. If we plot x + y = 5 and 2x + 2y = 10, the lines are:

(A) Parallel (B) Coincident
(C) Intersecting (D) Perpendicular

Q13. 5 pencils and 7 pens cost Rs 50. 7 pencils and 5 pens cost Rs 46. Cost of one pencil is:

(A) Rs 3 (B) Rs 5
(C) Rs 4 (D) Rs 6

Q14. If x = 1, y = 2 is a solution of 4x – 3y + k = 0, then k =

(A) 2 (B) –2
(C) 5 (D) –5

Q15. The lines 5x + 7y = 35 and 5x + 7y = 50 are:

(A) Coincident (B) Parallel
(C) Intersecting (D) Perpendicular

Q16. For what value of k will the system kx + 3y = k – 3, 12x + ky = k have no solution?

(A) k = 6 (B) k = –6
(C) k = 6 or k = –6 (D) k = 0

Q17. A boat goes 24 km upstream and 36 km downstream in 6 hours. Also it goes 36 km upstream and 24 km downstream in 6.5 hours. Speed of stream is:

(A) 2 km/h (B) 4 km/h
(C) 6 km/h (D) 8 km/h

Q18. A and B together can complete a piece of work in 12 days. A alone takes 30 days. Number of days B alone takes is:

(A) 20 days (B) 18 days
(C) 24 days (D) 15 days

Q19. The sum of digits of a two-digit number is 8. When the digits are reversed, the number becomes 18 less than the original. Original number is:

(A) 53 (B) 26
(C) 44 (D) 62

Q20. If x + 2y = 8 and 2x + ky = 16, find k for infinite solutions.

(A) 2 (B) 4
(C) 6 (D) 8

Q21. Aman is 4 times as old as his daughter. 5 years ago, he was 7 times as old. Present age of Aman is:

(A) 40 yr (B) 32 yr
(C) 48 yr (D) 36 yr

Q22. A railway half-ticket costs half the full fare, but reservation charges are the same on a half-ticket as on a full ticket. One reserved first-class ticket costs Rs 2530, and one full and one half reserved ticket cost Rs 3810. The full fare is:

(A) Rs 2500 (B) Rs 2400
(C) Rs 2520 (D) Rs 2480

Q23. 5 women and 4 men can together finish an embroidery in 4 days, while 4 women and 6 men can finish it in 3 days. Time taken by 1 woman alone is:

(A) 18 days (B) 36 days
(C) 12 days (D) 24 days

Q24. Find x, y if 47x + 31y = 63 and 31x + 47y = 15.

(A) x = 2, y = –1 (B) x = 1, y = 2
(C) x = –1, y = 2 (D) x = 3, y = –2

Q25. The area of triangle formed by the lines x = 0, y = 0 and 3x + 4y = 12 is:

(A) 6 sq units (B) 12 sq units
(C) 24 sq units (D) 5 sq units

— End of Worksheet —